

From Hints to Textbooks

Studying Scaffolds through LLMs and Bayesian Methods

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Roadmap

- 1 The dataset and the two taxonomies
- 2 Method and infrastructure
- 3 Headline 1: counting & probability is different
- 4 Headline 2: masterpieces are unlike everything else
- 5 My textbook vs. MATH number theory
- 6 Working with LLMs on the book
- 7 Takeaways

The dataset: SocraticMath

Where it comes from

- Competition problems from the MATH benchmark^a, each augmented with a *chain of Socratic questions*^b
- One JSON file per problem: `problem`, `level` (1–5), `solution`, `socratic_questions`

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Scale

- 7 domains $\times$ 565–590 problems = **4,043 problems**
- **31,613 Socratic hints** (≈ 7.8 per problem)

^aHendrycks et al., *Measuring Mathematical Problem Solving with the MATH Dataset*, NeurIPS 2021.

^bDing et al., *Boosting LLMs with Socratic Method for Conversational Mathematics Teaching*, CIKM 2024.

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- One JSON file per problem: problem, level (1–5), solution, type, socratic_questions

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domain	problems
algebra	576
counting & probability	584
geometry	575
intermediate algebra	565
number theory	583
prealgebra	590
precalculus	570
total	4,043

Taxonomy 1: four hint types

Goldin, Koedinger & Aleven (2013) — plus one extra type for Socratic chains:

[f] **feature-pointing** draws attention to a specific aspect of *this* problem

“Notice that these two angles are vertical angles”

[r] **principle-stating** supplies a general rule, definition, or theorem

“Vertical angles are equal in measure”

[b] **bottom-out** directly advances the solution — set up, substitute, solve

“Add these two known angles to find the missing one”

[e] **extension** pushes past the current problem — applications, generalizations

“Can you think of other situations where this matters?”

Goal: learn a 4×4 **transition matrix** M per group, where p_{ij} is the probability the next hint is type j given the current one is type i .

A worked example (algebra hint_0)

Let $f(x) = \begin{cases} ax + 3 & x > 2 \\ x - 5 & -2 \leq x \leq 2 \\ 2x - b & x < -2 \end{cases}$ Find $a + b$ if f is continuous.

- 1 [f] Can you explain what a piecewise function is and how it is represented algebraically?
- 2 [r] How would you determine if the function is continuous where different pieces meet?
- 3 [f] What conditions need to be satisfied for a piecewise function to be continuous?
- 4 [b] Can you identify the value of a and b that would make the function continuous?
- 5 [b] How would you approach solving for a and b ?
- 6 [b] Are there any restrictions or constraints on the values of a and b ?
- 7 [e] Can you explain the significance of continuity in real-life applications?
- 8 [e] Can you think of other examples where continuity is important?

Labels shown are my hand labels. On this chain [Claude reproduces all 8/8](#)

Taxonomy 2: Quarfoot's nine problem types

A musical metaphor, ordered (roughly) least → most mathematical substance:

	type	what it is
FN	First Notes	one-step, single-path, algorithmic; fluency
AC	Accidentals	looks routine, hides a twist; targets a misconception
CH	Chords	one idea, several steps; deepens a single concept
ET	Etudes	drills one technique; word-problem dressing
IM	Improvisations	a playground; first attempts fail, so you experiment
IN	Interpretations	wide-open solution space; compare qualitatively different paths
FP	First Pieces	first problems needing multiple ideas plus a macro-level plan
SP	Showpieces	very hard, hinges on a flash of insight; cognitive
MP	Masterpieces	many skills converging on a generalizable truth; capstones

The nine types, illustrated from my number theory textbook

Claude's classification of *Olympiad Number Theory* problems (533 total), one example each:

type	problem	source
FN	Find x such that $1 + x + x^2 + x^3 + \cdots = 4$.	Mandelbrot
AC	Show there are no integers x, y with $1691x + 1349y = 1$.	Ch. 5
CH	The sum of 18 consecutive positive integers is a perfect square. Find the smallest possible such sum.	AMC 12
ET	Compute $\sum_{m=1}^{75} (2 + 3m)$.	Ch. 1
IM	Pages numbered $1-n$; one page number was accidentally added twice, giving 1986. Which page?	AIME
IN	Prove these Fibonacci identities <i>in two different ways</i> .	Ch. 2
FP	Find the 8th term of 1440, 1716, 1848, $\dots$, formed by multiplying corresponding terms of two arithmetic sequences.	AIME
SP	Evaluate $\prod_{n=2}^{\infty} \frac{n^3-1}{n^3+1}$.	Putnam
MP	Prove every positive integer is a sum of non-consecutive Fibonacci numbers.	Zeckendorf

Method: one Claude call per problem, two classifications

- `claude-opus-4-8` receives the problem, difficulty level, worked solution, and the numbered Socratic chain, and returns strict JSON:
 - ① one hint label (`f/r/b/e`) per question, in order
 - ② one Quarfoot type for the whole problem
- Prompt contains both taxonomies' definitions + the worked example above as the single few-shot anchor
- Every response is **validated** (label count = question count, legal codes only) and retried on failure — batch retry, then per-problem retry
- Downstream analysis: group the chains (by domain, or by problem type), count transitions, Dirichlet-multinomial posterior per group, row-averaged symmetrized KL, similarity e^{-KL}
- Artifacts: one JSON per problem in `classifications_llms/`, figures in `outputs_llms/`

Infrastructure: written by Claude, run on Claude

Built with Claude Code

- Agent wrote the backend, classifier, CLI commands, and plots inside the existing `uv` project
- Backend auto-selects: Anthropic SDK if an API key is set, else headless `claude -p` calls on my Claude subscription
- Batched 10 problems/call (≈ 410 calls), 12 parallel workers

Robustness (needed it!)

- Resumable: each problem's JSON written on arrival; re-runs skip finished work
- Circuit breaker: 3 consecutive all-fail batches $\Rightarrow$ stop (usage limit), auto-resume later

Compute & cost

- Throughput ≈ 53 problems/min $\Rightarrow \approx$ **80 min of active compute** for all 4,043 problems (≈ 3 h wall clock: hit the subscription usage limit twice, auto-resumed)
- Marginal cost: **\$0** on the subscription ($\approx$ \$60 API-equivalent at Opus list prices)

It starts with staring at the raw transition matrices

Per-domain 4×4 matrices over **[f][r][b][e]**. Six of the seven look roughly like algebra's. One doesn't:

$$M_{\text{algebra}} = \begin{pmatrix} \mathbf{0.227} & 0.224 & 0.498 & 0.051 \\ 0.160 & 0.272 & 0.502 & 0.067 \\ 0.089 & 0.158 & 0.600 & 0.153 \\ 0.030 & 0.073 & \mathbf{0.193} & \mathbf{0.704} \end{pmatrix} \quad M_{\text{count. \& prob.}} = \begin{pmatrix} \mathbf{0.418} & 0.126 & 0.421 & 0.035 \\ 0.184 & 0.242 & 0.441 & 0.133 \\ 0.070 & 0.104 & 0.715 & 0.111 \\ 0.030 & 0.060 & \mathbf{0.079} & \mathbf{0.831} \end{pmatrix}$$

- $f \rightarrow f = 0.418$: the highest of *any* domain (runner-up: prealgebra 0.331) and nearly $2\times$ algebra's 0.227 — c&p hints chain observation $\rightarrow$ observation
- $e \rightarrow b = 0.079$: the lowest of any domain — once a c&p chain reaches extensions it almost never drops back into computation ($e \rightarrow e = 0.831$)
- The other rows are unremarkable — the signal is concentrated, not diffuse

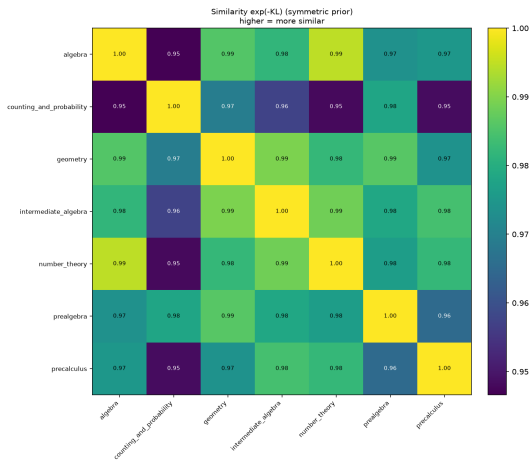
Compressing a comparison into one number

Count transitions, add the all-ones Dirichlet prior, normalize rows — e.g. the counting & probability f -row has counts (491, 147, 495, 40), so $p_{ff} = \frac{491+1}{1173+4} \approx 0.418$. Then compare two domains row by row: symmetrized KL, averaged over the four rows.

row	f	r	b	e	mean
$\frac{1}{2}[KL(A\ C) + KL(C\ A)]$	0.096	0.030	0.030	0.063	0.0549

Similarity = $e^{-0.0549} \approx 0.947$ — the *lowest* of any domain pair. The f -row does the most work, exactly the observation-chaining row from the previous slide.

Across all 21 pairs: counting & probability is the outlier



- The dark stripe is counting & probability's row/column: mean similarity to the other domains **0.958**, the lowest of any domain
- Geometry — the domain you might have guessed — is among the most *central* (mean 0.981)
- One-vs-rest permutation test: c&p diverges most from the pooled rest — sym. KL 0.034, $\approx 1.7\times$ the runner-up; Holm-adjusted $p < 0.001$

Why counting & probability reads differently

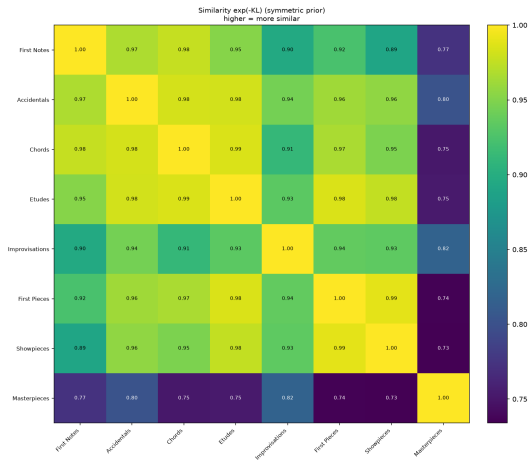
- Its hint *marginals* already stand out — most feature-pointing, least principle-stating of any domain:

	f	r	b	e
counting & probability	23.9%	13.9%	50.0%	12.1%
all domains (mean)	18.9%	18.6%	51.5%	10.9%

- And its *dynamics*: hints stack observations ($f \rightarrow f$ at 0.418) before computing, and once a chain reaches extensions it stays there ($e \rightarrow e$ at 0.831 vs. 0.704 for algebra)
- Intuition: “notice how many ways...” beats “recall the theorem...” — combinatorics hints scaffold by *observation*, not by *principle*

Takeaway: the outlier domain isn’t the one with the strangest notation — it’s the one whose scaffolding *strategy* differs.

Same machinery, new grouping: transitions by problem type



- Group the same 4,043 chains by Claude's *Quarfoot type* instead of by domain, and recompute the similarity matrix
- Every pair of non-masterpiece types: similarity ≥ 0.89
- Masterpieces vs. everything: **0.73–0.82** (mean 0.77)

Domains differ by a few percent.
Masterpieces sit *outside the family* — the largest effect anywhere in this study.

What makes a masterpiece different?

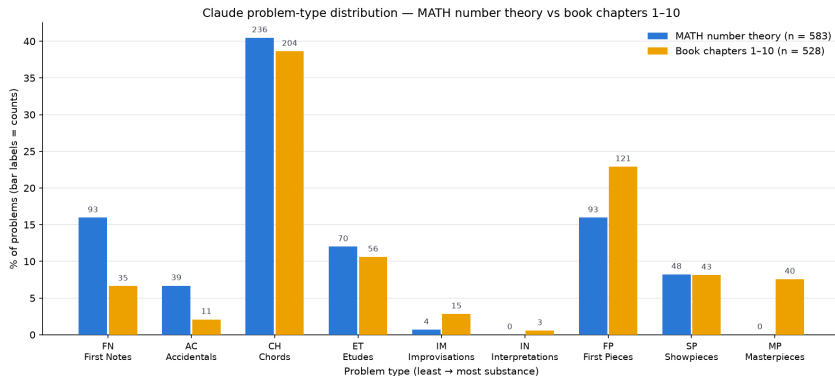
Raw transition counts, all masterpiece chains:

	$\rightarrow f$	$\rightarrow r$	$\rightarrow b$	$\rightarrow e$
f	0	1	1	0
r	0	0	1	0
b	0	0	5	2
e	0	0	0	0

Caveat: that is 2 chains, 12 hints — Claude finds only **2 masterpieces in all 4,043 MATH problems**. Read the direction, not the digits.

- **The extension coda vanishes.** Every other type's chains settle into extensions at the end ($e \rightarrow e$ from 0.63 to 0.83); masterpiece chains never repeat an extension — *the problem is the extension*
- **Bottom-out carries the chain:** 7 of 10 transitions start at b — the hints construct the result step by step
- **The endpoint of the substance axis.** Moving $FN \rightarrow SP$, $b \rightarrow b$ climbs 0.56 $\rightarrow$ 0.78 and the principle-stating share falls 31% $\rightarrow$ 13%; masterpieces push both further
- **And they're vanishingly rare here** — so rare a competition item bank can't even estimate their dynamics. Where *do* they live?

Same classifier, two corpora: MATH number theory vs. book ch. 1–10



- The *middles* match: Chords 40.5% vs. 38.6%; Etudes and Showpieces within ~ 1.5 points
- The *tails* diverge in opposite directions

Reading the tails as pedagogy

MATH number theory (n = 583)

- Thick floor of fluency checks: $FN + AC = 22.6\%$ (“what is the units digit of...”)
- **Zero Masterpieces** — headline 2 in corpus form
- A competition *item bank*

Book chapters 1–10 (n = 528)

- Light on drills: $FN + AC = 8.7\%$
- More multi-idea planning: FP 22.9% vs. 16.0%
- **7.6% Masterpieces** — culminating capstones (Zeckendorf, USAMO, Putnam)
- A curated *learning arc*

The arc is visible *within* the book too: the share of high-substance problems (FP/SP/MP) climbs from **24%** in Ch. 1 to $\approx$ **50%** by Ch. 9–10.

Why this is credible: identical classifier, prompt, and rubric on both corpora — near-identical middles, sharply different tails $\Rightarrow$ properties of the corpora, not classifier noise.

What this offers people who write textbooks

- **An audit tool for your problem mix.** Type profiles make the shape of a book quantitative: drills vs. concept-deepeners vs. capstones, per chapter and overall — and comparable against item banks or other texts.
- **A check on the arc.** You can verify the substance axis actually climbs the way you intended (mine does: 24% → 50% high-substance).
- **Gap-finding.** What I learned about my own book:
 - essentially no *Interpretations* (3 of 533) — almost nowhere do I ask students to solve one problem *two different ways* and compare
 - Improvisations are rare too (15) — few low-stakes “playground” problems
 - the capstones I hoped were there really do register (40 Masterpieces)
- **For tutoring systems:** hint scaffolding should adapt per domain — combinatorics wants observation-chaining, algebra wants principle-then-compute.

Working with LLMs on the book: observations

Working with LLMs on the book: observations (cont.)

Takeaways, caveats, next steps

Takeaways

- 1 Counting & probability scaffolds by *observation*, not principle — the clear domain outlier, with the mechanism visible in the raw matrices
- 2 Masterpieces' hint dynamics look like no other type's — and MATH barely contains them (2 of 4,043); capstones live in curated texts, not item banks
- 3 Problem-type profiles act as corpus fingerprints: item bank vs. curated arc — and the whole pipeline is cheap enough to be routine (≈ 80 min, \$0 marginal)

Caveats

- No human-labeled evaluation set yet — the earlier “gold set” was itself LLM-authored, so I dropped it; a real one is the top priority
- The masterpiece transition matrix rests on 2 chains — the outlier direction is consistent, but the estimate is prior-dominated
- Claude saw the worked solution: single-solution write-ups may bias it against open-ended types (IM/IN) and Masterpieces; one model, one prompt — no inter-rater κ yet

Next steps: human gold set + Cohen's κ with a second annotator; generate Socratic chains for the book's 40 Masterpieces to test headline 2 at real sample size; hint transitions vs. difficulty level; HMM over noisy labels.

Thanks! Questions?

Code available [here](#).